HW #9

1)

2) ~~K~~

$W = -[(2\cdot 10) + (1\cdot 10)\frac{1}{2} - (1\cdot 10)\frac{1}{2} + (2\cdot 10)\frac{1}{2}]$

$= -[10J] = 10J$

b) $W = K_f - K_i$

$K_f = W + K_i$

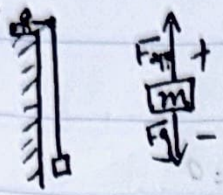
$= W + \frac{1}{2}mv_i^2$

$= 10J + \frac{1}{2}\cdot 250kg\cdot(20.0 m/s)^2$

$= 10J + 50J$

$= 60J$

2)



Given: $F_g = mg$, $m = 12.0\,kg$, $W = 4000\,J$

$x_f = 0$, $V = k$

$$W = \vec{F}_{app} \cdot \vec{d}$$

$$\vec{d} = \frac{W}{F_{app}}$$

$$= \frac{W}{mg}$$

$$= \frac{4000\,J}{12\,kg \cdot -9.81\,N}$$

$$= 33.98\,m$$

$$\vec{F}_{net} = 0$$

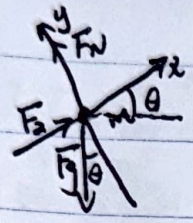
$$= \vec{F}_{app} + \vec{F}_g$$

$$0 = \vec{F}_{app} + \vec{F}_g$$

$$-\vec{F}_g = \vec{F}_{app}$$

$$-mg = \vec{F}_{app}$$

3) Given: $x_0 = 0$, $x_f = 1.8m$, $m = 14kg$, $\theta = 10°$, $V_0 = .5 m/s$, $V_f = 1.5 m/s$, $F_{app} = K$



a) $W_g = \cancel{F_g d}\ F_g d \cos\theta$

$= mg\, d \cos\theta$

$= 14kg \cdot 9.81 m/s^2 \cdot \cos 100°$

$= -23.85J$

b) $\cancel{W} = \Delta K = K_f - K_i$

$= \frac{1}{2} m (V_f^2 - V_i^2)$

$= \frac{1}{2} \cdot 14kg (1.5 m/s^2 - .5 m/s^2)$

$= 14J$

c) $W = \Delta K$

$= 14J$

d) $W = W_g + W_{app}$

$W_{app} = W - W_g$

$= 14J - -23.85J$

$= 37.85J$

5)

Given: $m = 2.0\,kg$, $F_x = +6.0x\,N$

$F = ma$

$= m\frac{V}{t}$

$6 \cdot 3 = 2 \cdot \frac{8}{t}$

$\frac{18}{16} = \frac{1}{t}$

$t = \frac{8}{9}\,s$

② $F = m\frac{V}{t}$

$6 \cdot 4 = 2 \cdot \frac{V}{8/9}$

$18 \cdot \frac{8}{9} = V$

$16 = V$

ⓑ $F = m\frac{V}{t}$

$6x = 2 \cdot \frac{7}{8/9}$

$3x = 7 \cdot \frac{9}{8}$

$x = \frac{7}{1} \cdot \frac{9}{8} \cdot \frac{1}{3}$

$= \frac{21}{8}\,m$

6) Given: $F = 58000\,N$, $V_{max} = 1002\,km/h$

$\because F = m\frac{d}{t^2}$

$V = \frac{d}{t}$

$P = m\frac{d^2}{t^3}$

$\therefore P = FV$

$F = 58000\,kg\frac{m}{s^2}$

$V = 1002\,km/h \cdot \frac{1000\,m}{1\,km}$

$= 1002000\,m/h \cdot \frac{1\,h}{3600\,s}$

$= 278.3\,m/s$

$P = FV$

$= 58000\,kg\frac{m}{s^2} \cdot 278.3\frac{m}{s}$

$= 1.61414 \times 10^7\,W$

$P = 1.61414 \times 10^7\,W \times \frac{1\,W}{746\,hp}$

$= 21637\,hp$

7) Given: $\mathbb{P}$, $\mathbb{P} = k$, m, t

$$W = \Delta K = \frac{1}{2} m V^2 \qquad a = \frac{dV}{dt}$$

$$\mathbb{P} = \frac{dW}{dt} \qquad V = \frac{dx}{dt}$$

$$= \frac{d}{dt}\left(\frac{1}{2} m V^2\right)$$

$$= mV \frac{dV}{dt}$$

$$\cancel{= mVa}$$

$$\frac{\mathbb{P}}{m} = V \frac{dV}{dt}$$

$$\frac{\mathbb{P}}{m} dt = V dV$$

$$\int \frac{\mathbb{P}}{m} dt = \int V dV$$

$$\frac{\mathbb{P}}{m} t = \frac{1}{2} V^2$$

$$V^2 = 2\frac{\mathbb{P}t}{m}$$

$$V = \sqrt{2\frac{\mathbb{P}t}{m}}$$

$$\frac{dx}{dt} = \sqrt{2\frac{\mathbb{P}t}{m}}$$

$$dx = \left(2\frac{\mathbb{P}t}{m}\right)^{1/2} dt$$

$$\int dx = \int \left(2\frac{\mathbb{P}t}{m}\right)^{1/2} dt$$

$$x = \left(2\frac{\mathbb{P}}{m}\right)^{1/2} \int t^{1/2} dt$$

$$= \left(2\frac{\mathbb{P}}{m}\right)^{1/2} \frac{2}{3} t^{3/2}$$

$$s = x = \left(\frac{8}{9}\frac{\mathbb{P}}{m}\right)^{1/2} t^{3/2}$$

8) $\int_{x_i}^{x_f} F(x)\, dx$

$$= \int_0^{16.7} k_1 x + k_2 x^3 \, dx$$

$$= \int_0^{16.7} 204x + -.233x^3 \, dx$$

$$= 102x^2 - .05825x^4 \Big|_0^{16.7}$$

$$= 28446.78 - 4530.66$$

$$= 23916.12 \text{ J}$$